

Beepul Bharti

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Research Interests

- Safe anytime-valid inference: sequential testing, e-processes, and online learning
- Theory of responsible AI: interpretability, explainability, and algorithmic fairness
- Uncertainty quantification: calibration, conformal prediction, and related methods
- ML for high-stakes decision-making, e.g., in governance and healthcare

Education

- 2020– **Johns Hopkins University**
Ph.D. Biomedical Engineering
Dissertation: Statistical Foundations for Trustworthy Machine Learning
Advisor: Jeremias Sulam
Committee: Anqi Liu, Carey Priebe, and Jessica Sorrell
- 2016–20 **Duke University**
B.A. Mathematics & B.S. Biomedical Engineering
Honors: Cum Laude, Departmental Distinction

Experience

- Sum' 2024 **Genentech**, Machine Learning Research Intern San Francisco, CA
Deep Learning Theory and Algorithms Lab (DELTA)
- Explainability for motif discovery in genomic neural networks
- Sum' 2017 **Duke University**, Bass Connections Fellow Durham, NC
Department of Biostatistics
- Machine learning to predict schizophrenia-related hospital admission

Publications (*Indicates Equal Contribution)

- 2026 Parameter-Free and Group Conditional Online Conformal Prediction
Beepul Bharti, Ambar Pal, Jacopo Teneggi, Jeremias Sulam
Under review.
- 2026 Global Sequential Testing for Multi-Stream Auditing
Beepul Bharti, Ambar Pal, Jeremias Sulam
Under review.
- 2026 Let Me Explain, Again: Multiplicity in Local Sufficient Explanations
Ryan Pilgrim, Beepul Bharti, Jeremias Sulam, Rene Vidal
Transactions on Machine Learning Research (TMLR).
- 2025 Multiaccuracy and Multicalibration via Proxy Groups
Beepul Bharti, Mary Versa Clemens-Sewall, Paul Yi, Jeremias Sulam
International Conference on Machine Learning (ICML).
- 2025 Uncovering BioLOGICAL Motifs and Syntax via Sufficiency and Necessary Explanations
Beepul Bharti, Gabrielle Scalia, Alex Tseng
ICLR Workshop on Machine Learning for Genomics Explorations.

- 2024 Sufficient and Necessary Explanations (and What Lies in Between)
Beepul Bharti, Paul Yi, Jeremias Sulam
Conference on Parsimony and Learning (CPAL). **Oral**.
- 2024 Evaluation of AI Algorithmic Biases in Radiology
Paul H. Yi, Preetham Bachina, Beepul Bharti, Sean P. Garin, Adway Kanhere, Pranav Kulkarni, David Li, Vishwa S. Parekh, Samantha M. Santomartino, Linda Moy, Jeremias Sulam
Radiology.
- 2023 Estimating and Controlling for Equalized Odds via Sensitive Attribute Predictors
Beepul Bharti, Paul Yi, Jeremias Sulam
Advances in Neural Information Processing Systems (NeurIPS).
- 2023 SHAP-XRT: The Shapley Value Meets Conditional Independence Testing
Jacopo Teneggi*, Beepul Bharti*, Yaniv Romano, Jeremias Sulam
Transactions on Machine Learning Research (TMLR).
- 2023 Sex Imbalance Produces Biased Deep Learning Models for Knee Osteoarthritis Detection
David Li, Beepul Bharti, Jinchi Wei, Jeremias Sulam, Paul H. Yi
Canadian Association of Radiologists Journal.

Honors & Awards

- 2023 Alpha Eta Mu Beta: National Biomedical Engineering Honor Society
- 2022 JHU Mathematical Institute for Data Science Fellowship
- 2020 JHU Institute of Computational Medicine Fellowship
- 2016–2020 Duke University Dean's List
- 2019–2020 Duke University Pratt Fellowship
- 2019 Tau Beta Pi: The Engineering Honor Society

Teaching

Teaching Assistant

- 2023 EN.580.69: Biomedical Data Design, JHU
- 2021 EN.580.697: Computational Cardiology, JHU
- 2018 ECE 110L: Fundamentals of Electrical and Computer Engineering, Duke

Presentations

Talks

- 2026 Global Sequential Testing for Multi-Stream Auditing
UPenn CS Theory Seminar
- 2025 Sufficient and Necessary Explanations (and What Lies in Between)
Second Conference on Parsimony and Learning
- Sufficient vs. Necessary Explanations
Machine Learning in Healthcare Club, University of New South Wales
- 2024 SHAP-XRT: The Shapley Value Meets Conditional Independence Testing
Explainable AI Seminars at Imperial College London
- Algorithmic Fairness in Machine Learning and Data Science
EN.540.464: Advanced Biomedical Data Science for Biomedical Engineering
- 2023 Evaluating Fairness of AI Models in Radiology
Radiological Society of North America (RSNA) Annual Meeting

Fairness in Machine Learning
EN.540.464: Advanced Biomedical Data Science for Biomedical Engineering

2022 Shapley Values and Hypothesis Testing
QMUL Intelligent Sensing Winter School

Posters

2026 Parameter-Free and Group Conditional Online Conformal Prediction
Safe Anytime-Valid Inference (SAVI) Conference

2026 Global Sequential Testing for Multi-Stream Auditing
Safe Anytime-Valid Inference (SAVI) Conference

2024 Certifying Fairness with Incomplete Sensitive Information
SIAM Conference on Mathematics of Data Science

2023 Fairness via Sensitive Attribute Predictors
Columbia University Workshop on Fairness in Operations and AI

Fairness with Missing Sensitive Attributes
Johns Hopkins AI-X Foundry Fall Symposium

Shapley Values and Hypothesis Testing
Bern Interpretable AI Symposium

2018 Using Machine Learning to Predict Schizophrenia-Related Hospital Admission
Duke School of Medicine Clinical Research Day

Service

2025 Foreign Exchange Student Mentor
JHU Foreign Exchange Student Program

2024 High School Student Mentor
JHU Whiting School of Internships in Science and Engineering (WISE)

2023 Mentor Liaison
JHU Biomedical Engineering Application Assistance Program (BMEAAP)

Reviewing

Conferences: ICLR, ICML, FAccT, ISIT, AISTATS, NeurIPS, CPAL, and COLM
Journals: TMLR and Medical Image Analysis